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Solving operating room scheduling problems using large language model based multi-agent architecture

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Efficient operating room scheduling is a complex task due to the diversity of surgical types, high patient demand, and limited hospital resources. Since OR operations account for a substantial share of hospital costs, effective scheduling solutions are critical. Large language model (LLM)-based multi-agent systems offer new opportunities to support optimization for this problem. This study evaluates an LLM-based multi-agent framework for generating and refining optimization code for OR scheduling. Leveraging iterative collaboration among specialized agents, the proposed iterative multi-agent (IMA) framework consistently outperformed both single-agent and traditional multi-agent baselines. Across 10 LLMs and 35 scenarios, IMA achieved an average feasible schedule rate of 82.91%, compared with 65.59% and 52.53% for the baselines. Experiments on 1050 synthetic cases and three real OR datasets further showed that IMA-generated code produced solutions close to those of expert-written mixed-integer programming models, while a case study of hospital as-is schedules demonstrated improvements in OR utilization and overtime reduction. These results suggest that LLM-based multi-agent systems can provide a practical and effective approach for OR scheduling optimization.

Surgical services are among the most significant contributors to both hospital revenue and operational costs. They require extensive investments in human and physical resources, including operating rooms (ORs), surgical teams, and medical equipment. These resources are typically fixed and long-term, making efficient utilization critical. Ineffective scheduling can lead to underutilized operating rooms, increased financial waste as each unused minute in one OR costs an estimated $15 to $20, and decreased satisfaction among both patients and surgeons[1–3]. As such, OR scheduling is vital to balancing financial sustainability with quality care delivery[4,5].

Hospitals typically generate a preliminary operating room schedule for elective procedures about 1 week in advance, with final time slots confirmed the day before surgery. This process involves balancing limited resources with patient needs, often under time and capacity constraints. Despite this structured approach, operating room scheduling remains inherently complex. It involves coordinating multiple interdependent decisions, including selecting procedures, assigning operating rooms and staff, determining start times, and accommodating both elective and emergency patients[6].

Recent advances in artificial intelligence (AI), particularly in large language models (LLMs), have created new opportunities across various areas of healthcare. These models have been applied to tasks such as disease classification and detection[7,8], clinical decision support[9], patient assistance[10], operating room management[3], and admission prediction in emergency settings[11]. They have also shown potential for improving hospital resource allocation, including operating room scheduling and capacity planning[11–13]. Compared with traditional heuristic or statistical methods, which often rely on manual rule design and substantial effort[14–18], LLMs offer flexible, scalable, and data-driven alternatives for healthcare operations. They can support complex planning, prioritization, and scheduling tasks with greater efficiency[19,20]. Moreover, LLMs enable easier cross-institution deployment by focusing on clinical meaning rather than strict data formats. Unlike traditional models that rely on standardized inputs, LLMs can interpret diverse documentation styles, allowing them to generalize across different electronic health record systems (EHRs)[21]. As healthcare systems face increasing demands for personalized and efficient care delivery, LLMs are

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well-positioned to enhance decision-making by effectively balancing clinical needs with operational constraints[22]. Recent studies have introduced LLM-based multi-agent frameworks designed to exceed the capabilities of a single LLM[23,24]. These systems consist of multiple autonomous agents that collaborate through planning, discussion, and decision-making, much like human teams. A common approach involves role-playing, where each agent takes on a specific role to collectively solve complex tasks[25]. For example, multi-agent systems have been applied in software development to support collaborative programming processes[26–28] and in document understanding to handle retrieval and reasoning over long or multi-modal content[29,30]. In addition, multi-agent architectures have been studied as an approach to improve performance on complex reasoning tasks through coordinated interaction among agents[31,32]. General-purpose platforms such as OpenClaw further demonstrate the practical viability of deploying multiple autonomous agents in real-world automation environments[33]. Although LLM-based multi-agents have demonstrated potential across various non-medical domains, their application and evaluation within healthcare remain limited[34].

Based on this background, we propose a two-stage operating room scheduling framework developed using an LLM-based multi-agent architecture. Our main contributions are summarized as follows:

- We formulate the target problem as a two-stage operating room scheduling task. In the first stage, patients are assigned to operating days and rooms under resource and efficiency constraints. In the second stage, a priority-aware heuristic determines the sequence of surgeries within each day.
- We design an iterative multi-agent architecture in which specialized agents collaborate to interpret clinical requirements, generate optimization code, and revise it in response to feedback, thereby improving code-generation reliability and the feasibility of the resulting schedules.
- We evaluate the framework across 35 synthetic scenarios and three real-world hospital datasets. The experiments compare IMA with single-agent and traditional multi-agent baselines as a code generation and correction framework, assess the resulting solutions against expert-written mixed-integer programming models, and use a case study based on as-is hospital schedules to illustrate potential operational impact in practice.

Results

Experimental evaluation

To assess the effectiveness of our proposed IMA framework, we conducted comparative experiments against two baseline approaches: single-agent (SA) framework and multi-agent (MA) framework. In the SA approach, a single agent performs the task: it interprets the input, develops the mathematical model, generates the optimization code, and evaluates the solution, all in a single step. This approach does not include the iterative error-correction mechanism seen in the IMA framework. In the MA framework, multiple agents are used in parallel, similar to IMA, but without the iterative process. Each agent performs a single task, such as problem formulation, model creation, or solution evaluation, and their actions do not iterate or correct based on feedback from other agents. This makes the MA framework less flexible than the IMA approach, in which agents interact and iteratively refine their outputs. For consistency across trials, the maximum number of attempts for IMA was set to three in all experiments. Ablation studies were performed to isolate the contributions of the iterative error-correction mechanism and the multi-agent structure. The IMA framework was further tested on real-world operating room scheduling data involving 40,615 patients. Model performance was evaluated across three distinct clinical scenarios, focusing on task completion rate and solution accuracy. Moreover, we compared the outputs with those from a Gurobi optimization model manually developed by operations research experts.

Performance of frameworks and LLMs

Table 1 reports the feasible schedule rates of the three frameworks: SA, MA, and IMA, evaluated across different difficulty levels (simple, normal, and hard) using ten LLMs: GPT-3.5-Turbo, GPT-4o, GPT-5.2, Gemini-2.5, Gemini-3-pro, Claude-3.5, Claude-4, Llama-4, DeepSeek-R1, and Qwen3. A scenario was counted as successful if the generated optimization code executed successfully and produced a feasible OR schedule that satisfied all modeled constraints within the predefined maximum number of attempts. The feasible schedule rate was then calculated as the proportion of successful scenarios within each difficulty group. Across all settings, IMA consistently achieved the highest feasible schedule rates. With GPT-5.2, IMA produced feasible schedules for all scenarios, reaching 100% in simple, normal, and hard cases. Other models, such as GPT-4o, DeepSeek-R1, and Gemini-3-pro, also demonstrated strong performance with IMA, with feasible schedule rates above 90% in hard scenarios.

Among the LLMs, GPT-5.2 enabled IMA to produce a feasible schedule for every scenario, achieving a perfect 100% feasible schedule rate in simple, normal, and hard cases, as shown in Fig. 1. In contrast, GPT-3.5-Turbo, as illustrated in Fig. 2, exhibited substantially lower feasible schedule

Table 1 | Results of SA, MA, and IMA across different difficulty levels and LLMs

Model	Simple scenario (count = 7)				Normal scenario (count = 17)				Hard scenario (count = 11)			
	SA	MA	IMA	Avg. Iter. (IMA)	SA	MA	IMA	Avg. Iter. (IMA)	SA	MA	IMA	Avg. Iter. (IMA)
	Feasible schedule rate				Feasible schedule rate				Feasible schedule rate			
GPT 3.5-TURBO	14.29%	28.57%	42.86%	2.85	11.76%	17.65%	47.06%	2.88	0.00%	0.00%	36.36%	3.36
GPT 4o	71.43%	85.71%	100.00%	1.14	35.29%	82.35%	100.00%	1.24	36.36%	81.82%	100.00%	1.45
GPT 5.2	85.71%	85.71%	**100.00%**	**1.14**	88.24%	88.24%	**100.00%**	**1.17**	81.82%	90.91%	**100.00%**	**1.18**
Gemini-2.5	71.43%	71.43%	85.71%	1.57	52.94%	76.47%	94.12%	1.53	54.55%	63.64%	90.91%	1.63
Gemini-3-pro	85.71%	85.71%	100.00%	1.14	94.12%	88.24%	100.00%	1.23	100.00%	81.82%	100.00%	1.27
Claude-3.5	57.14%	71.43%	85.71%	1.71	47.06%	58.82%	82.35%	2.00	45.45%	54.55%	63.64%	2.18
Claude-4	57.14%	85.71%	85.71%	1.28	58.82%	70.59%	88.24%	1.52	54.55%	72.73%	90.91%	1.64
Llama-4	71.43%	71.43%	85.71%	1.57	29.41%	47.06%	76.47%	2.17	54.55%	72.73%	90.91%	1.63
DeepSeek-R1	71.43%	85.71%	100.00%	1.14	35.29%	82.35%	94.12%	1.41	54.55%	81.82%	90.91%	1.45
Qwen3	28.57%	42.86%	57.14%	2.57	17.65%	23.53%	52.94%	2.70	9.09%	18.18%	45.45%	3.18
Avg.	61.43%	71.43%	84.29%	1.61	47.06%	63.53%	83.53%	1.79	49.09%	61.82%	80.91%	1.90

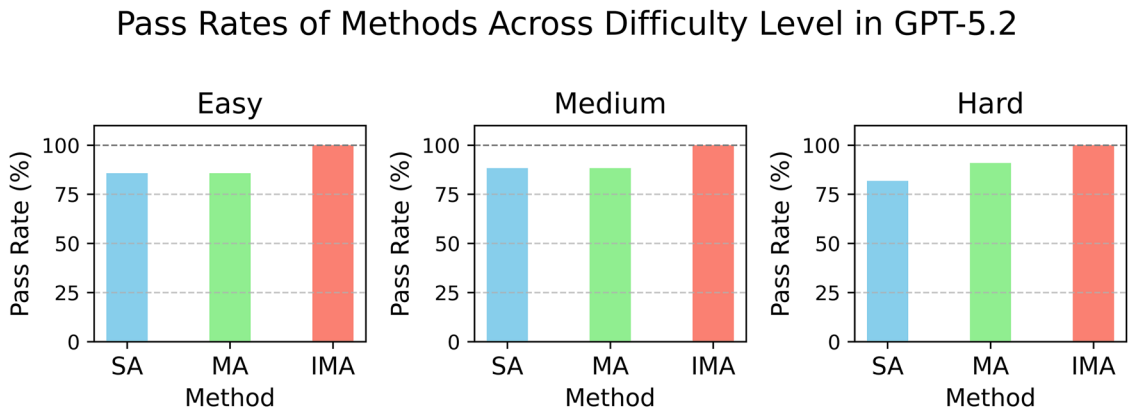


Fig. 1 | Framework comparison by difficulty level (GPT-5.2).

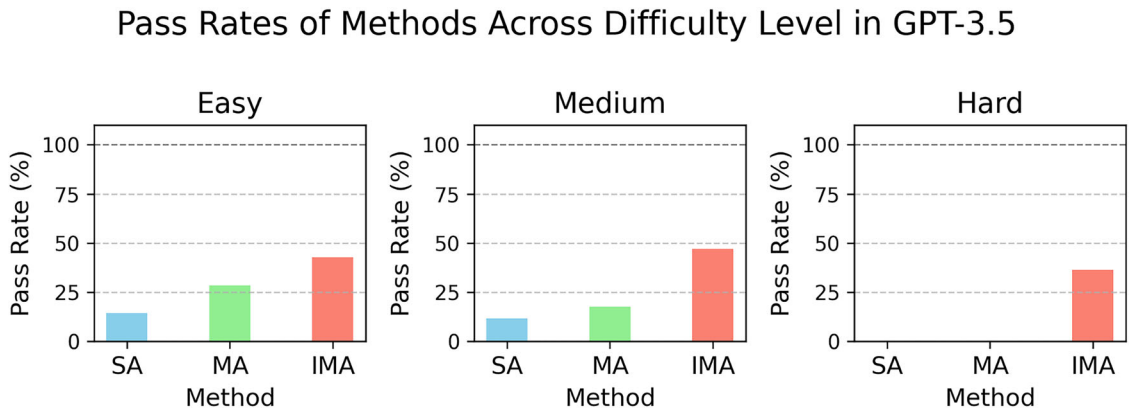


Fig. 2 | Framework comparison by difficulty level (GPT-3.5).

rates across all frameworks and difficulty levels, with IMA producing feasible schedules for only 42.86% of simple scenarios and 36.36% of hard scenarios. Both SA and MA performed even worse, particularly in hard cases, where they often failed to return feasible schedules. Models such as Claude-3.5 and Llama-4 demonstrated moderate performance, with IMA feasible schedule rates ranging from 63% to 91% in hard scenarios. The Avg. Iter. (IMA) column indicates how many rounds of revision and re-testing IMA needed on average before producing a feasible schedule. Lower values indicate that the framework reached a valid schedule more efficiently, whereas unsuccessful cases were counted using the maximum number of iterations. GPT-5.2 required the fewest iterations, with an average of 1.14 iterations across all difficulty levels. Other models, such as Gemini-3-pro and DeepSeek-R1, required slightly more iterations, with averages of 1.23 and 1.41, respectively. In contrast, models such as GPT-3.5-Turbo and Qwen3 showed relatively poor performance and required more iterations to reach feasible schedules. In summary, IMA consistently outperformed the other frameworks, demonstrating both higher feasible schedule rates and greater efficiency in terms of iteration count across different models and difficulty levels.

IMA scalability and solution quality

Table 2 summarizes the performance of the IMA across scenarios of varying difficulty: simple, normal, and hard, and three problem sizes: 30, 50, and 100 patients. The maximum allowed solution time was set to 300s. For simple scenarios, the framework solved all 70 test cases at each size with zero optimality gap. The average solution time increased from 0.05 s for 30 patients to 28.21 s for 100 patients, demonstrating near-linear scalability. In normal scenarios, feasibility remained around 94% across all problem sizes, with the average optimality gap consistently below 1%. Solution time increased from 1.72 s to 58.93 s as the number of patients grew. For hard scenarios, feasibility was maintained at 100%, though the average optimality gap rose from 0.43% at 30 patients to 3.06% at 100 patients. The solution time for the largest instances reached 113.41 s.

Ablation study of RAG

To quantify the contribution of RAG, we conduct an ablation study within the IMA framework while restricting each agent to a single execution cycle. This configuration removes iterative refinement and ensures that performance differences reflect the direct impact of retrieval augmentation. The evaluation includes three real-world scenarios, each comprising 20 weekly cases sampled from nine clinical departments. As reported in Table 3, each scenario contains 20 test cases, and incorporating RAG consistently improves the feasible solution rate across all three scenarios. With RAG enabled, the success rates are 100%, 95%, and 90% for Scenarios 0, 1, and 2, respectively. Without RAG, these values decline to 60%, 80%, and 40%, representing reductions of 40, 15, and 50 percentage points. The performance gap is most pronounced in Scenario 2, where the feasible solution rate more than doubles with RAG. In terms of computational cost, enabling RAG results in a slight increase in average runtime across scenarios. However, this increase is modest relative to the substantial improvement in reliability. These results demonstrate that retrieval augmentation delivers measurable gains in robustness with limited additional overhead.

Comparative analysis with human-written algorithms on real data and benchmark datasets

To evaluate the effectiveness of the IMA-generated algorithm, we compared its performance with expert-written mixed-integer programming models solved with the same Gurobi optimizer. The key metric was total cost,

Table 2 | Evaluation of IMA on scenarios with varying difficulty levels and test sizes

Difficulty level	Pass scenarios count	Patient = 30				Patient = 50				Patient = 100			
		Test case count	Feasible solution rate	Avg. gap (%)	Avg. Time (s)	Test case count	Feasible solution rate	Avg. gap (%)	Avg. time (s)	Test case count	Feasible solution rate	Avg. gap (%)	Avg. time (s)
Simple	7	70	100%	0%	0.05	70	100%	0%	4.85	70	100%	0.13%	28.21
Normal	17	170	98.82%	0.00%	1.72	170	94.12%	0.15%	7.49	170	94.12%	0.88%	58.93
Hard	11	110	100%	0.43%	6.96	110	100%	0.57%	9.47	110	100%	3.06%	113.41

defined as the sum of operating room usage fees: each room incurred a base cost of 100 units per day with a standard 10-h usage limit, and any overtime was penalized at 20 units per additional hour. Each case corresponded to 1 week of scheduling data from a hospital department. Statistical comparisons between the two approaches were conducted using paired *t*-tests and Wilcoxon signed-rank tests.

We applied the evaluation to three real-life OR scheduling datasets. The first two were collected from Ghent University Hospital (UZGent) in Belgium in 2021, including eight sub-datasets (G_1–G_8)[35] and two sub-datasets (G_00 and G_01)[36]. The original data format of both (G_1–G_8) and (G_00, G_01) contains patient IDs and the corresponding surgery start and end times, but does not specify emergency patient types. Therefore, in our preprocessing, 20% of the patients were randomly designated as emergency cases. As shown in Table 4, results across G_1–G_8 indicate that the IMA method achieves outcomes almost identical to those produced by human-written models, with most *p*-values close to 1, suggesting no significant differences. Table 5 further confirms this finding: for G_00 and G_01, the IMA method consistently matched human solutions, with only minor variations in certain cases.

We also tested the framework on datasets from multiple departments of West China Hospital, as illustrated in the green box of Fig. 3. Table 6 summarizes the results. In Neurosurgery, no significant differences were observed across all scenarios, with *t*-test *p*-values above 0.3 and Wilcoxon *p*-values at or above 0.05. In Ophthalmology, the total costs from both methods were identical across all scenarios, yielding *t*-test values of 1.00 and Wilcoxon *p*-values of 1.00, indicating complete agreement. Similar patterns were observed in ENT, Orthopedics, Cardiac, and Urology departments, where cost outcomes were identical, or differences were statistically insignificant with *p*-values greater than or equal to 0.06. For ENDO, Plastics, and Pediatrics, minor numerical differences were detected in some scenarios, but the corresponding *p*-values from both the *t*-test and the Wilcoxon test generally exceeded 0.16, indicating no statistically meaningful gap. Overall, these findings indicate that the IMA algorithm delivers cost outcomes equivalent to those of manually written optimization logic, regardless of clinical department, case complexity, or patient volume. This consistency highlights the potential of the IMA framework as a robust and scalable solution for automated operating room scheduling.

As-is versus LLM-based OR scheduling

Table 7 compare the as-is schedules generated by the hospital's current scheduling system with the LLM-based optimized schedules across nine departments. For each department, we used 16 weeks of elective surgery data. Emergency and ICU cases were excluded because their scheduling is often shaped by substantial human intervention and case-specific factors, which could reduce the accuracy of a direct comparison. The table reports three operational indicators for both scheduling approaches: total OR days, OR utilization rate, and OR scheduling overtime rate. Total OR-days represents the number of operating-room day units used to accommodate the weekly workload, where one OR-day refers to one operating room scheduled on one calendar day. OR utilization rate measures how fully the allocated OR time is used. OR scheduling overtime rate represents the proportion of OR time scheduled outside regular operating hours, namely before 8:00 and after 18:00. Several clear patterns emerge from the comparison. First, the LLM-based schedules reduce total OR days in most departments, and the Wilcoxon test shows statistically significant differences in nearly all cases. Second, OR utilization under the LLM-based schedules is more concentrated, generally ranging from 99% to 107%, whereas the as-is schedules show much larger variation across departments. This indicates that the optimized schedules use OR capacity more evenly and keep utilization closer to the planned capacity. Third, OR scheduling overtime rates are substantially lower under the LLM-based schedules in all departments, with consistently significant *p*-values. For example, overtime decreases from 25.30% to 3.01% in Neurosurgery and from 12.51% to 1.01% in ENDO.

Table 3 | Effect of RAG on solution robustness and runtime in the IMA framework

Scenario ID	Case count	With RAG		Without RAG	
		Feasible cases *n*/*N* (%)	Avg. runtime (s)	Feasible cases *n*/*N* (%)	Avg. runtime (s)
0	20	20/20 (100%)	36.34	12/20 (60%)	31.90
1	20	19/20 (95%)	35.60	16/20 (80%)	29.56
2	20	18/20 (90%)	36.53	8/20 (40%)	31.37

Table 4 | Comparison of algorithm performance between human-written and IMA methods (dataset from ref. 35)

Department	Scenario ID	Case count	Avg operation duration (min)	Total patient num	Human		IMA		*T*-test *p*-value	Wilcoxon *p*-value
					Total cost					
					Median	Mean	Median	Mean		
G_1	1	20	86	1472	1200.00	1179.00	1200.00	1178.95	0.19	0.18
	2	20	86	1472	1200.00	1200.00	1200.00	1200.00	1.00	1.00
	3	18	85	1330	1241.50	1286.33	1241.50	1286.33	1.00	1.00
G_2	1	20	86	1480	1200.00	1182.33	1200.00	1182.10	0.00	0.00
	2	20	86	1480	1200.00	1215.00	1200.00	1215.00	1.00	1.00
	3	20	86	1480	1216.00	1223.70	1216.00	1223.70	1.00	1.00
G_3	1	20	90	1510	1300.00	1261.83	1300.00	1261.75	0.06	0.06
	2	20	90	1510	1300.00	1290.00	1300.00	1290.00	1.00	1.00
	3	20	90	1510	1325.00	1315.58	1325.00	1315.58	1.00	1.00
G_4	1	20	87	1519	1200.00	1221.67	1200.00	1221.50	0.01	0.02
	2	20	87	1519	1200.00	1235.00	1200.00	1235.00	1.00	1.00
	3	20	87	1519	1288.00	1270.05	1288.00	1270.05	1.00	1.00
G_5	1	20	85	1478	1121.67	1162.83	1121.00	1162.60	0.00	0.01
	2	20	85	1478	1200.00	1200.00	1200.00	1200.00	1.00	1.00
	3	20	85	1478	1200.00	1217.05	1200.00	1217.05	1.00	1.00
G_6	1	20	87	1484	1200.00	1195.83	1200.00	1195.65	0.01	0.02
	2	20	87	1484	1200.00	1220.00	1200.00	1220.00	1.00	1.00
	3	20	87	1484	1240.50	1243.95	1240.50	1243.95	1.00	1.00
G_7	1	20	85	1490	1176.67	1185.33	1176.00	1185.10	0.00	0.01
	2	20	85	1490	1200.00	1200.00	1200.00	1200.00	1.00	1.00
	3	20	85	1490	1200.00	1225.45	1200.00	1225.45	1.00	1.00
G_8	1	20	88	1494	1213.33	1213.67	1213.00	1213.45	0.00	0.01
	2	20	88	1494	1300.00	1255.00	1300.00	1255.00	1.00	1.00
	3	20	88	1494	1300.00	1270.10	1300.00	1270.10	1.00	1.00

"Total cost" denotes a unit-based cost and does not represent a monetary value in any specific currency.

Table 5 | Comparison of algorithm performance between human-written and IMA methods (dataset from ref. 36)

Department	Scenario ID	Case count	Avg operation duration (min)	Total patient num	Human		IMA		*T*-test *p*-value	Wilcoxon *p*-value
					Total cost					
					Median	Mean	Median	Mean		
G_00	1	24	94	1370	1025.00	1003.06	1024.50	1002.88	0.00	0.01
	2	24	94	1370	1100.00	1029.17	1100.00	1029.17	1.00	1.00
	3	24	94	1370	1095.50	1038.17	1095.50	1038.17	1.00	1.00
G_01	1	24	100	2544	1923.33	1926.11	1923.00	1925.96	0.01	0.02
	2	24	100	2544	2000.00	1954.17	2000.00	1954.17	1.00	1.00
	3	24	100	2544	2000.00	1994.04	2000.00	1994.04	1.00	1.00

"Total cost" denotes a unit-based cost and does not represent a monetary value in any specific currency.

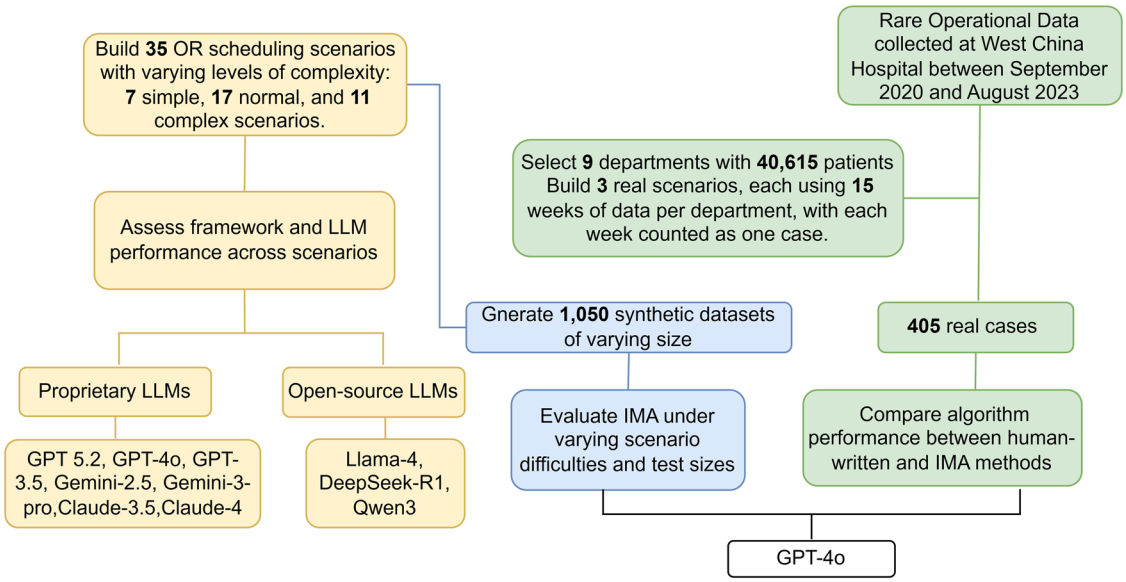


Fig. 3 | Flowchart of data selection and testing. The testing process is divided into three parts, each indicated by a different color. The orange section shows the evaluation of seven LLMs across different frameworks and 35 scenarios. The blue section represents the testing of algorithms generated by IMA on datasets of varying sizes. Finally, the green section illustrates the testing process using real-world scenario data.

Cardiac surgery exhibits a distinct pattern compared with the other departments. The LLM-based schedule increases the mean weekly OR-days from 59.75 to 106.44; however, the as-is schedule has an exceptionally high OR utilization rate of 177.28%, indicating substantial overtime and a highly concentrated workload. Under the LLM-based schedule, OR utilization decreases to 101.50%, while the OR scheduling overtime rate declines sharply from 48.49% to 3.05%. These findings suggest that the additional OR-days allow the surgical workload to be distributed more evenly and substantially reduce excessive overtime, rather than indicating lower scheduling efficiency. Overall, the LLM-based schedules promote a more balanced allocation of OR workload across departments while effectively limiting excessive overtime.

Interpretability of the correction process

Table 8 provides a more explicit view of how the multi-agent workflow detects, routes, and corrects errors across the 35 OR scheduling test scenarios. For each LLM, the table reports three aspects of the correction process: the number of IMA self-corrections, the agent ultimately responsible for correction, and the type of error involved. The responsible agent is determined by the Project Manager based on the error report returned by the Tester. Error types are grouped into four categories: Solution process, Model issues, Implementation issues, and Solver API use issues. The results show that both correction responsibility and error type vary markedly across LLMs. Responsibility for correction is shared between the OR expert and the programmer, although the split varies substantially by model, with an overall average of 50% for each role. Across all IMA corrections, Solution process and Model issues account for 34.88% and 32.56%, respectively, indicating that the main challenge lies less in low-level syntax errors and more in maintaining valid optimization logic, execution flow, and formulation quality.

Different LLMs also exhibit distinct error profiles. For example, Gemini-2.5 is corrected almost entirely by the programmer, and 85.71% of its IMA corrections are associated with issues related to Solver API use, suggesting failures concentrated at the implementation interface level. In contrast, Llama-4 relies more heavily on the OR expert, and 77.78% of its corrections are associated with Model issues, indicating a failure mode more related to formulation quality than coding details. Overall, this analysis shows that LLM failures are model-specific and that the correction records provide a more interpretable view of where and how different models break down during scheduling.

Discussion

We introduced an IMA framework based on LLM to tackle the complexities of operating room scheduling. To evaluate its effectiveness, we created 35 scheduling scenarios with varying difficulty and tested it alongside two baselines: SA and MA, using ten language models: GPT-3.5-Turbo, GPT-4o, GPT-5.2, Gemini-2.5, Gemini-3-pro, Claude-3.5, Claude-4, Llama-4, DeepSeek-R1, and Qwen3. The results reveal two key findings. First, IMA's iterative and collaborative design significantly outperforms both SA and MA approaches, especially in more complex scenarios. Second, the choice of LLMs has a substantial impact on outcomes. Among the tested LLMs, GPT-5.2 consistently demonstrated superior reasoning and coordination capabilities, particularly in complex scenarios, while GPT-3.5 and other models showed lower success rates. These findings suggest that combining the IMA framework with stronger LLMs can substantially improve the reliability of generating executable optimization code and feasible scheduling solutions for operating room scheduling problems.

Beyond overall performance, the stability and scalability of the framework are also important for practical deployment. To further evaluate these aspects, we generated 30 datasets of varying sizes for each of the 35 scenarios. Results indicate that IMA consistently produces high-quality, feasible solutions, even as problem size and complexity increase. Although larger and more complex datasets naturally demand more computational resources, the framework maintained robust and predictable performance across all test cases. These findings confirm that the IMA framework not only delivers performance on par with expert-written optimization logic but does so consistently across clinical departments, case complexities, and patient volumes. We further evaluated IMA using three real surgical datasets, comparing it against expert-written algorithms. Statistical tests showed no significant cost differences, confirming that IMA matches expert performance across departments and scales.

From a deployment perspective, the monetary cost of LLM inference remains manageable. Using the case study dataset and GPT-5.2, priced at $1.75 per million input tokens and $14 per million output tokens[37], the estimated cost per scheduling instance was approximately $0.0506, based on an average of 4,768 input tokens and 3,015 output tokens per case. This highlights a practical trade-off between cost and capability. Lower cost

Table 6 | Comparison of algorithm performance between human-written and IMA methods across departments and scenarios

Department	Scenario ID	Case count	Avg operation duration (min)	Total patient num	Human		IMA		*T*-test *p*-value	Wilcoxon *p*-value
					Total cost					
					Median	Mean	Median	Mean		
Neurosurgery	1	15	214	161	338.56	152.82	338.56	152.82	0.43	1.00
	2	15	284	951	2510.91	556.72	2511.01	556.72	0.33	0.05
	3	15	274	1976	4891.10	4997.27	4891.40	4999.13	0.40	0.88
Ophthalmology	1	15	28	3141	1136.17	1030.48	1136.17	1030.48	1.00	1.00
	2	15	28	2788	1000.00	980.00	1000.00	980.00	1.00	1.00
	3	15	29	2915	1154.67	1068.38	1154.67	1068.38	1.00	1.00
ENT	1	15	114	1544	3100.00	2486.67	3100.00	2426.67	0.06	0.07
	2	15	114	1436	2100.00	2286.67	2100.00	2286.67	1.00	1.00
	3	15	118	1432	2100.00	2340.00	2100.00	2340.00	1.00	1.00
Orthopedics	1	15	182	2059	6300.00	4866.67	6300.00	4866.67	1.00	1.00
	2	15	186	2006	6500.00	4880.00	6500.00	4880.00	1.00	1.00
	3	15	181	1514	2700.00	3560.00	2700.00	3560.00	1.00	1.00
ENDO	1	15	167	606	1300.00	1248.91	1300.00	1244.98	0.17	0.18
	2	15	170	636	1400.00	1333.33	1400.00	1326.67	0.33	0.32
	3	15	169	527	1300.00	1101.80	1300.00	1093.33	0.18	0.18
Plastics	1	15	183	280	100.00	440.00	100.00	440.00	0.17	0.18
	2	15	176	360	500.00	800.00	500.00	786.67	0.16	0.16
	3	15	182	387	700.00	880.00	700.00	873.33	0.33	0.32
Urology	1	15	131	1368	1800.00	2513.33	1800.00	2513.33	1.00	1.00
	2	15	130	1258	1400.00	2280.00	1400.00	2280.00	1.00	1.00
	3	15	132	1254	1800.00	2353.33	1800.00	2353.33	1.00	1.00
Pediatrics	1	15	122	1738	3500.00	3146.67	3500.00	3134.40	0.33	0.32
	2	15	122	1598	3500.00	2913.33	3500.00	2913.33	1.00	1.00
	3	15	122	1782	3600.00	3213.33	3600.00	3213.33	1.00	1.00
Cardiac	1	15	241	2228	4200.00	3680.00	4200.00	3680.00	1.00	1.00
	2	15	243	2663	4900.00	4240.00	4900.00	4240.00	1.00	1.00
	3	15	234	2007	4200.00	3373.33	4200.00	3373.33	1.00	1.00

"Total cost" denotes a unit-based cost and does not represent a monetary value in any specific currency.

models may be sufficient for routine or lower complexity scheduling tasks, whereas stronger models may be more appropriate when the scheduling environment is more constrained, uncertain, or operationally critical. At the hospital level, the optimized schedules may also generate meaningful operational savings. As shown in Table 7, across nine departments over a 16-week period, total OR demand decreased from 405.39 to 392.00 OR-days, corresponding to a reduction of 13.39 OR-days, while the average OR utilization rate increased from 84.47% to 100.98%, and the average OR scheduling overtime rate fell from 15.60% to 2.21%. Using the reported average operating cost of approximately USD 1709 per OR-day for a tertiary Grade A hospital[38], this reduction corresponds to an estimated saving of about USD 22,797 over 16 weeks. This estimate is conservative, as it covers only nine departments over roughly one quarter and does not include additional labor-related costs, overtime staffing burden, scheduling coordination effort, or potential savings that could arise if the same scheduling approach were extended to other campuses within the hospital system. Together, these results suggest that the framework supports flexible deployment strategies under different budget constraints while also offering the potential for meaningful practical savings. The case study from West China Hospital also provides evidence on the practical feasibility of deploying the IMA framework in a large tertiary setting. In this study, IMA was evaluated across multiple OR scheduling scenarios with different levels of complexity and was further tested using real surgical data from one of the largest hospitals in southwestern China. This environment offers a realistic setting for assessing not only the quality of optimization but also whether the framework can operate effectively in a hospital where specialties differ substantially in caseload, case mix, and scheduling pressure. The results suggest that the framework can support scheduling in environments with heterogeneous departments, variable workloads, and substantial operational complexity.

The potential value of the framework may also extend beyond OR scheduling. Its core architecture does not rely on a single specialty-specific rule set, but rather on the interaction among task decomposition, error detection, role-based correction, and optimization refinement. For this reason, it may be adaptable to other hospital scheduling problems that also require balancing resource constraints, clinical priorities, and operational rules, such as ICU bed allocation, staff scheduling, outpatient procedure scheduling, and other capacity-constrained planning tasks. These problems share several structural features with OR scheduling, including limited resources, competing demands, dynamic arrivals, and the need to coordinate clinical and operational objectives within a formal decision process. At the same time, the extent of this generalizability should be interpreted cautiously. Problems such as ICU bed management and emergency-related scheduling often involve more frequent real-time changes, greater dependence on clinician judgment, and objectives that are harder to formalize than those in elective OR planning. As a result, the overall workflow may be transferable, but some components, especially those related to model

Table 7 | Comparison between the as-is solution and the LLM-based optimized solution (scenario ID = 1)

Department	*n* weeks	Mean weekly cases	Mean weekly surgery hours	Total OR-days			OR utilization rate (%)			OR scheduling overtime rate (%)		
				As-is solution mean	LLM-based optimized mean	Wilcoxon *p*	As-is solution means	LLM-based optimized mean	Wilcoxon *p*	As-is solution means	LLM-based optimized mean	Wilcoxon *p*
Neurosurgery	16	171.00	794.36	84.19	80.44	0.02	96.12	101.67	0.02	25.30	3.01	0.00
Ophthalmology	16	377.00	289.15	53.94	37.31	0.00	68.31	100.26	0.00	5.62	0.51	0.00
ENT	16	12.94	15.87	6.00	1.81	0.00	29.55	107.41	0.00	0.17	9.09	0.01
Orthopedics	16	192.06	620.39	67.94	65.12	0.05	93.75	99.99	0.02	18.21	1.04	0.00
ENDO	16	44.44	127.55	16.25	13.62	0.00	81.25	99.71	0.00	12.51	1.01	0.00
Plastics	16	43.62	135.70	20.38	14.44	0.00	68.29	99.11	0.00	5.06	1.30	0.01
Urology	16	144.56	351.61	50.75	38.00	0.00	72.02	99.54	0.00	10.07	0.31	0.00
Pediatrics	16	134.31	322.34	46.19	34.81	0.00	73.63	99.61	0.00	14.97	0.53	0.00
Cardiac	16	206.81	1055.53	59.75	106.44	0.00	177.28	101.50	0.00	48.49	3.05	0.00
Aggregate	–	**1326.74**	**3712.50**	**405.39**	**392.00**	–	**84.47**	**100.98**	–	**15.60**	**2.21**	–

The Aggregate row reports sums of mean weekly cases, mean weekly surgery hours, and OR-days, as well as unweighted department-level means for OR utilization and scheduling overtime rates.

formulation, objective design, and real-time updating, would likely require further adaptation. Therefore, while the present results suggest that the IMA framework has potential as a broader hospital scheduling approach, stronger claims of transferability will require validation on additional datasets, institutions, and decision contexts.

In clinical practice, the IMA framework should be positioned as a decision-support tool rather than an autonomous decision-maker. A more realistic implementation model is human AI collaboration, in which the system generates feasible scheduling recommendations while clinicians or operational staff retain responsibility for defining key priorities, such as patient urgency, sequencing rules, and department-specific requirements. This approach may improve scheduling efficiency while preserving clinical oversight, which is essential for both safety and accountability. Finally, for such a workflow to be usable in routine care, practical deployment conditions and privacy requirements must also be considered. In hospital settings with limited information technology resources, operational burden depends not only on runtime, but also on inference cost, the number of model interactions, and the need for repeated correction as scheduling complexity increases. These demands may be moderated by reusable institutional knowledge resources, more standardized electronic health records, and clearer data-processing pipelines. At the same time, patient privacy must remain a central requirement for clinical adoption. Sensitive information should not be directly exposed during model interaction. Measures such as encryption, de-identification, and local execution within hospital-controlled environments may help reduce privacy risk while preserving the practical value of the framework.

Overall, this study demonstrates that LLMs can generate operating room scheduling solutions with accuracy and efficiency comparable to those produced through manual modeling and solving. By leveraging real-world hospital data and evaluating performance across diverse scenarios and leading LLMs, we show that the IMA framework consistently outperforms SA and MA approaches while maintaining scalability across both synthetic and real scheduling tasks. These findings highlight the potential of LLM-based decision support systems to improve operating room efficiency and reduce healthcare resource burdens. At the same time, several limitations should be acknowledged when interpreting these results. First, many surgical procedures involve inherent uncertainties in their duration. In real-world settings, such variability can significantly affect schedule feasibility and resource allocation. Incorporating stochastic elements or probabilistic estimates of surgical time into the modeling framework could enhance both the robustness and practical relevance of LLM-assisted solutions. Second, our experiments were limited to nine surgical departments. While the results demonstrate promising performance in these cases, they may not fully capture the diversity and complexity of operating room scheduling across different clinical specialties. Expanding the evaluation to include a broader range of departments, hospital settings, and case volumes will provide a more comprehensive understanding of the model's generalizability and effectiveness in varied real-world environments. Such validation is essential for building trust and facilitating integration into clinical workflows.

Future research should therefore focus on broadening the clinical applications of LLM-assisted operating room scheduling by incorporating dynamic, unpredictable factors such as emergency insertions and surgeon availability. While these uncertainties will improve the system's robustness and generalizability, they will also introduce added complexity. Currently, the agents in our system struggle to manage such intricate and variable scenarios effectively. To address this, we plan to take a more targeted approach by enhancing each agent individually. This will involve customizing methods for each agent, fine-tuning existing models, and exploring the integration of reinforcement learning (RL) alongside LLMs. By doing so, we aim to improve each agent's adaptability and decision-making

Table 8 | Error-handling patterns of different LLMs across 35 OR scheduling test scenarios

Model	IMA self-corrections count	Agent responsible for the correction		Error category (as % of IMA corrections)			
		OR expert	Programmer	Solution process	Model issues	Implementation issues	Solver API use issues
GPT-3.5-TURBO	20	70.00%	30.00%	40.00%	50.00%	10.00%	0.00%
GPT-4o	6	50.00%	50.00%	33.33%	33.33%	16.67%	16.67%
GPT-5.2	4	25.00%	75.00%	25.00%	25.00%	50.00%	0.00%
Gemini-2.5	7	0.00%	100.00%	0.00%	0.00%	14.29%	85.71%
Gemini-3-pro	5	0.00%	100.00%	40.00%	0.00%	60.00%	0.00%
Claude-3.5	8	50.00%	50.00%	37.50%	12.50%	25.00%	25.00%
Claude-4	5	60.00%	40.00%	40.00%	40.00%	20.00%	0.00%
Llama-4	9	77.78%	22.22%	22.22%	77.78%	0.00%	0.00%
DeepSeek-R1	4	75.00%	25.00%	50.00%	25.00%	25.00%	0.00%
Qwen3	18	44.44%	55.56%	44.44%	22.22%	22.22%	11.11%
Average	**9**	**50.00%**	**50.00%**	**34.88%**	**32.56%**	**19.77%**	**12.79%**

Table 9 | Summary of surgical workload and case characteristics by department

Department	Daily surgeries (mean ± SD)	Max daily surgeries	Peak/mean (%)	Mean duration (min)	Emerg. cases (%)	Overtime cases (%)
Neurosurgery	29 ± 9	54	185	272	15	28
Ophthalmology	70 ± 14	96	137	46	5	6
ENT	25 ± 6	40	162	125	7	16
Orthopedics	42 ± 12	74	175	194	7	25
ENDO	10 ± 4	31	317	174	1	23
Plastics	9 ± 3	21	233	176	3	13
Urology	30 ± 9	55	186	148	5	20
Pediatrics	25 ± 8	46	183	130	11	15
Cardiac	24 ± 10	74	307	322	14	33

The departments listed here represent a selected subset of high-volume surgical services from West China Hospital and do not constitute an exhaustive list of all surgical specialties.

capabilities, ensuring they can navigate complex, real-world environments more effectively. These improvements are vital for maintaining the system's reliability across diverse and unpredictable clinical settings.

Methods

Setting and data collection

Our dataset consists of both synthetic and real-world operating room scheduling data. To evaluate the performance of the proposed Iterative Multi-Agent (IMA) framework, we first constructed 35 OR scheduling scenarios with varying levels of complexity: 7 simple, 17 normal, and 11 complex scenarios. Each scenario includes instances with varying patient volumes (30, 50, and 100 patients) to evaluate the robustness of the generated algorithms across problem sizes. For the synthetic datasets, surgery durations were randomly generated between 60 and 180 min to reflect realistic variability in surgical procedures. Each scenario includes 30 independent test instances. Details regarding scenario design and parameters are provided in the supplementary data.

We used three real-world operating room scheduling datasets: two publicly available datasets from Ghent University Hospital (UZ Gent) in Belgium[35,36], and one from West China Hospital, where we selected nine representative departments, classified as general surgery including Neurosurgery, Orthopedics, Urology, Cardiac and Plastics, internal medicine including Endocrinology, and other specialized services including Ophthalmology, ENT and Pediatrics. The portion of the study involving the West China Hospital dataset was conducted in accordance with the Declaration of Helsinki and was approved by the Biomedical Ethics Committee of West China Hospital, Sichuan University (Approval No. 2019-473; July 18, 2019). The approval remained valid throughout the study period. This selection reflects a wide range of surgical durations, patient volumes and resource requirements. For example, Neurosurgical procedures are typically long and involve a high proportion of emergency cases. While the average duration is relatively consistent across the department, operating room usage varies substantially, often leading to uneven resource allocation. In contrast, ophthalmic procedures are shorter and higher-volume. Approximately 65% are outpatient day surgeries, with cataract and refractive procedures being the most common. These surgeries require precise scheduling due to their compact duration and high throughput. Based on these departmental characteristics, we extracted representative surgical records to construct real-world scheduling scenarios. The raw operational data were collected at West China Hospital between September 2020 and August 2023. To reduce the potential impact of the COVID-19 pandemic on surgical activity, we selected 2023 records for analysis. During preprocessing, 87 records (0.21%) were excluded because of missing required fields or implausible numerical values. After preprocessing, 40,615 surgical records from nine departments were retained. Using these records, we constructed three real-world operating room scheduling scenarios by organizing the data into weekly

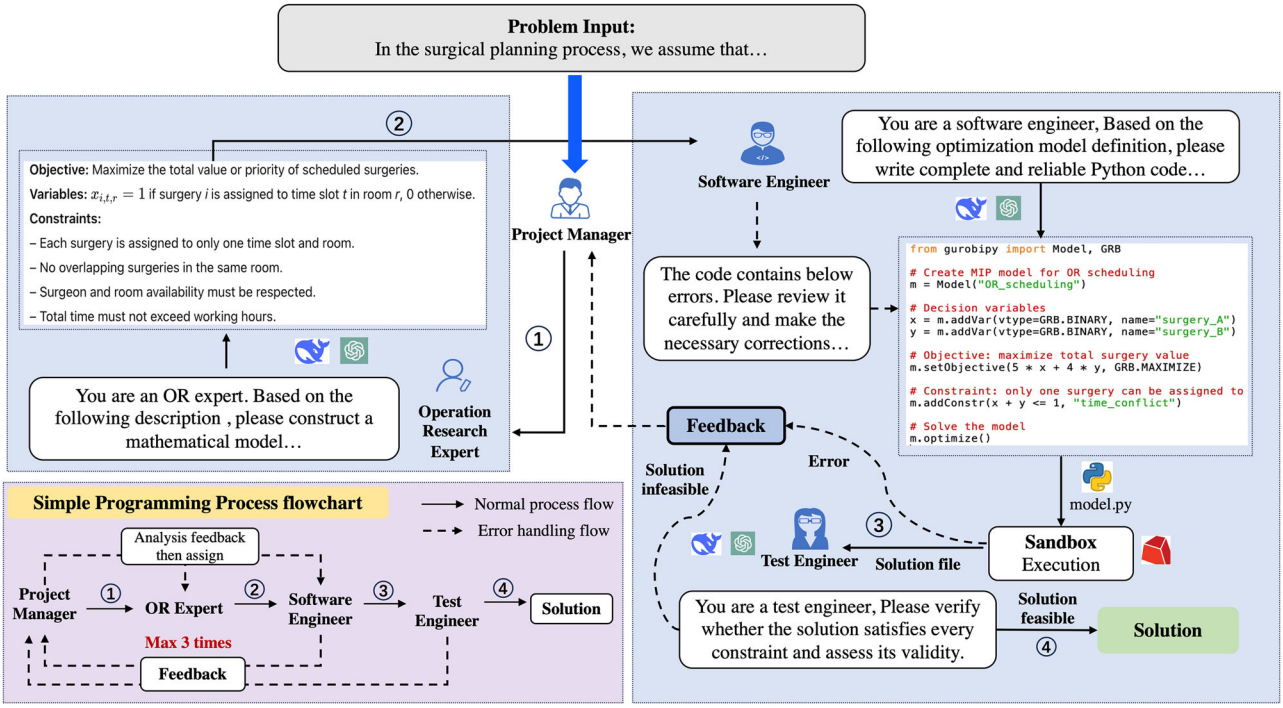


Fig. 4 | Illustrate the IMA framework for operating room scheduling, which coordinates four specialized LLM agents. Project Manager, Operations Research Expert, Software Engineer, and Test Engineer. The system iteratively interprets user input, formulates optimization models, generates and executes code, and validates results.

cases, with each department contributing 15 weeks of data and each week treated as an independent scheduling instance. The specific input settings for all scenarios are described in the data. Figure 3 illustrates the workflow of data selection and experimental evaluation. The testing process consists of three parts: evaluating seven LLMs across different frameworks and 35 scenarios, testing algorithms generated by the IMA framework on datasets of varying sizes, and validating the framework using real-world scenario data.

To further characterize the real-world setting, Table 9 summarizes surgical workload and case characteristics for a subset of high-volume surgical departments selected from the West China Hospital dataset. These departments were chosen to represent diverse workload patterns and operational characteristics rather than to provide an exhaustive classification of all surgical specialties. The reported metrics include average daily surgical volume, maximum daily workload, mean surgery duration, proportion of emergency cases, and percentage of surgeries requiring overtime. These statistics reveal substantial variation across departments. For example, the average daily surgical volume ranges from 9 cases in Plastics to 70 cases in Ophthalmology, while the proportion of emergency cases spans from 1% in Endocrinology to 15% in Neurosurgery. Surgery duration also differs markedly, from 46 min in Ophthalmology to 322 min in Cardiac surgery, directly affecting operating room occupancy and turnover efficiency. To further capture demand fluctuations, we calculated the peak-to-mean workload ratio, which measures the difference between the maximum and average daily workloads. This ratio highlights the instability of surgical demand across departments. Cardiac surgery and Endocrinology, for instance, show relatively large gaps between peak and average workloads, indicating stronger temporal variation in case demand, whereas departments with long and complex procedures, such as Neurosurgery, tend to exhibit higher overtime rates. These variations underscore the diverse operational challenges faced by different departments, offering a realistic foundation for assessing the robustness of the proposed scheduling framework.

Iterative multi-agent framework

We propose an IMA framework designed to solve operating room scheduling tasks through natural language interaction, as illustrated in Fig. 4. The coordination and logic of the multi-agent collaboration are detailed in Algorithm 1. The overall workflow follows a structured process in which user-provided instructions generate optimization code, which is then executed and evaluated iteratively.

The IMA framework includes four core agents, each assigned a distinct role in the optimization workflow: the Project Manager, the Operations Research Expert, the Software Engineer, and the Test Engineer. These agents can be implemented using either the same or different LLMs, depending on system design.

When a user submits a problem, the Project Manager first interprets the input and formulates a clear understanding of the task objectives, ensuring alignment with the intended scope. This interpretation is then passed to the Operations Research Expert, who is responsible for developing a precise mathematical model of the problem, including defining decision variables, constraints, and objective functions. Once the model is established, the Software Engineer translates the formulation into executable code and integrates it with the available dataset. The generated code is then run in a secure sandbox environment to ensure safety, reproducibility, and controlled resource usage. The optimization code is generated in Python and solved using the commercial mathematical optimization solver Gurobi, which is widely used to find high-quality solutions for complex decision-making problems. Following execution, the Test Engineer evaluates the output by verifying that the solution satisfies all defined constraints and performance criteria. If the code fails to run successfully or the solution violates any constraints, the system initiates a new iteration. In this cycle, the Project Manager reviews error logs, diagnoses the cause, and reassigns the task to the relevant agent. Each agent then refines revise their outputs accordingly. This iterative process continues until a valid and feasible solution is produced or the maximum number of iterations is reached.

Algorithm 1. Multi-agent collaboration

Agents: PM: Project Manager; OE: OR Expert; SE: Software Engineer; TE: Test Engineer
Input: pd_{text}: Problem description by user; T: Maximum number of attempts

$t \leftarrow 0$ ▷ **Initialize iteration counter**
$r \leftarrow$ "error" ▷ **Initialize result status**
while $r =$ "error" **and** $t < T$ **do**
  if $t = 0$ **then**
    ▷ **First round process**
    $PM_t^{\text{output}} \leftarrow \text{PM.generate}(pd_{\text{text}})$
    $OE_t^{\text{output}} \leftarrow \text{OE.model}(PM_t^{\text{output}})$
    $SE_t^{\text{output}} \leftarrow \text{SE.code}(OE_t^{\text{output}})$
  else
    ▷ **Subsequent rounds**
    $next_expert \leftarrow \text{PM.assignExpert}(el, TE_t^{\text{output}})$
    if $next_expert = OE$ **then**
      $OE_t^{\text{output}} \leftarrow \text{OE.remodel}(el, TE_t^{\text{output}}, OE_{t-1}^{\text{output}})$
      $SE_t^{\text{output}} \leftarrow \text{SE.recode}(OE_t^{\text{output}})$
    else
      $SE_t^{\text{output}} \leftarrow \text{SE.recode}(el, TE_t^{\text{output}}, SE_{t-1}^{\text{output}})$
    end if
  end if
  ▷ **Code execution and testing**
  $r,\ solution \leftarrow \text{executeInSandbox}(SE_t^{\text{output}})$
  $t \leftarrow t + 1$
end while
return $r, communication_log$

In this study, the maximum number of attempts was fixed at $T = 3$ based on preliminary experiments. We observed that most recoverable failures were resolved within three iterations, whereas allowing more attempts led to little additional improvement while increasing computational overhead. The workflow-level coordination cost of Algorithm 1 is linear in the maximum number of attempts T, because each iteration triggers a bounded sequence of agent actions. Therefore, the orchestration complexity is $O(T)$. In practice, the main computational burden comes from model inference, code execution, and optimization solving, whereas the coordination mechanism itself introduces only limited overhead.

Retrieval-augmented generation

Retrieval Augmented Generation (RAG) extends large language models by integrating external knowledge retrieval into the generation process. Instead of relying solely on pretrained parameters, RAG retrieves relevant information from a knowledge base and incorporates it into the prompt, enabling more accurate and context aware responses[39,40]. In OR scheduling, incorporating external knowledge is essential because decisions must satisfy complex and interdependent operational constraints. We therefore integrate RAG into our framework to leverage historical scheduling experience and support structured decision-making.

Figure 5 illustrates the overall workflow of the proposed IMA communication and RAG-based scheduling framework. The system consists of four agents: a Project Manager, an OR Expert, a Software Engineer, and a Tester. These agents collaborate through iterative dialog to develop and refine the OR scheduling model. In each round, the model is formulated, implemented, and evaluated. Based on validation feedback, the schedule is revised until it satisfies all constraints and passes the Tester's verification. Within this collaborative process, RAG provides structured guidance derived from historical scheduling experience. Past communication records, including problem descriptions, constraint formulations, objective definitions, and associated code, are stored in a searchable database. Each record is annotated using a manually predefined tag vocabulary, in which individual objectives and constraints are assigned corresponding tags according to deterministic rules. When a new scheduling task is received, candidate records are first retrieved based on semantic similarity, then filtered or re-ranked according to the overlap between their tags and the current task's objectives and constraints. The retrieved constraint patterns and code structures are subsequently incorporated into the prompt supplied to the language model. Rather than replicating prior solutions, the retrieved information serves as structured guidance to support constraint-compliant reasoning. This design grounds the agents' decisions in validated scheduling experience and improves the reliability and clinical applicability of the generated OR schedules. The contribution of RAG is further evaluated through ablation studies presented in the "Results" section.

Incorporating patient priority into operating room scheduling

In real-world operating room scheduling, a variety of patient-related factors must be considered. Table 10 summarizes key patient information commonly used to inform scheduling decisions, such as clinical urgency, physician preference, and patient characteristics. Information is collected from EHRs and quantified to enable scoring and ranking. For factors not recorded in the EHRs and not readily quantifiable, such as surgeon preference regarding surgery timing, information is obtained through preoperative consultation. To translate these scores into actionable scheduling decisions while respecting hospital resources and operating room availability, a surgery time window is defined for each patient, representing the earliest and latest permissible times for scheduling. Patients within this window are then scored and scheduled according to the quantified factors. If a patient has no specific timing preference, the time window defaults to the full working day. This integrated approach ensures that patient preferences are taken into account while maintaining overall scheduling feasibility and efficiency.

To address scheduling needs, we adopt a two-stage framework. In the first stage, the IMA framework assigns patients to specific days and operating rooms, taking into account constraints and overall costs. In the second stage, patients are sequenced using a heuristic algorithm based on a weighted priority score. For each patient i, the priority score S_i is computed as the sum of normalized patient-related factors f_{ij} multiplied by their respective weights w_j:

$$S_i = \sum_{j=1}^{n} w_j \tilde{f}_{ij}, \quad \tilde{f}_{ij} = \frac{f_{ij} - \min(f_j)}{\max(f_j) - \min(f_j)} \tag{1}$$

Patients are then ranked according to S_i, where higher scores reflect greater overall priority based on the weighted combination of patient-related factors. This allows emergencies and other high-priority cases to be scheduled earlier through a clear and efficient sequencing rule.

LLM configuration

To ensure a broad comparison, we tested several versions of selected proprietary models, covering both recent and older releases. We tested both proprietary (GPT-3.5[41], GPT-4o[42], GPT-5.2[37], Gemini-2.5[43], Gemini-3.1-Pro[44], Claude-3.5[45] and Claude-4[46]) and open-source LLMs (Llama-4[47], DeepSeek-R1[48], and Qwen3[49]) to support decision-making in operating room scheduling. For GPT-3.5, we used the `gpt-35-turbo-16k-0613`

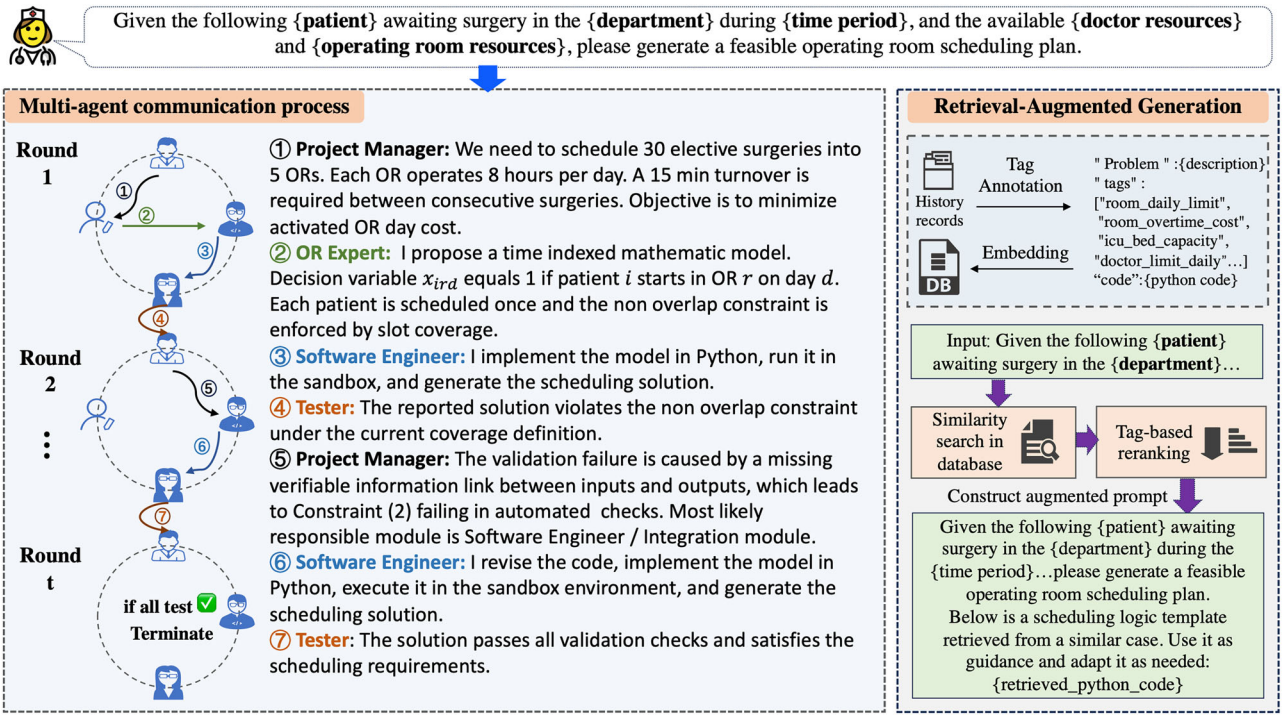


Fig. 5 | Overall workflow of the IMA communication and retrieval-augmented scheduling framework. The left panel shows the multi-agent communication process, where four agents iteratively develop and refine the OR scheduling solution until it passes all validation checks. The right panel illustrates the RAG module, which retrieves relevant historical cases through semantic search and tag-based reranking, and incorporates structured constraint information into the prompt to support feasible schedule generation.

Table 10 | Patient factors considered in operating room scheduling

Patient Factor	Description
Surgical urgency	Elective, urgent, or emergency procedures, determining scheduling priority.
ASA score	Reflects baseline physical status and perioperative risk.
Surgical classification	Complexity and invasiveness of the procedure affect duration and resource needs.
Wound classification	Infection risk category influences scheduling and postoperative planning.
Wound healing class	Healing potential or wound condition relevant to postoperative care.
Anesthesia type	Type of anesthesia planned, guiding OR setup and staffing.
Special populations	Age-specific considerations for neonates, children, and elderly patients.
Comorbidities	Existing conditions impacting surgical risk and scheduling.
Surgeon preference	Consideration of surgeon-specific requests or patient coordination.

model via the OpenAI API, setting the temperature to 0.2 to ensure deterministic outputs. Frequency and presence penalties were disabled to allow natural variation in responses. GPT-4o, configured with `gpt4o-mini-2024-07-18`, followed the same settings to enable fair comparison between model versions. GPT-5.2 was also invoked with similar settings to ensure consistency across proprietary models. Gemini-2.5-Pro and Gemini-3.1-Pro were run with a temperature of 0.2, top_p of 0.95, top_k of 64, and no penalties. Claude 3.5 Sonnet and Claude-4 used a similar configuration, relying on their default sampling parameters with temperature set to 0.2 and penalties disabled. For the open-source models, we used mostly default sampling settings to mirror the proprietary setups. For Llama-4-Maverick and DeepSeek-R1, we applied temperature 0.2, *top_p* 0.9, *top_k* 50, and no repetition penalties. Qwen3 was invoked with temperature 0.2, *top_p* 0.9, *top_k* 50, and penalties set to 0 for consistency across all open-source LLMs. These settings were chosen to ensure a fair comparison between model versions and to maintain consistency across experimental conditions.

Data availability

The open-source Ghent University Hospital dataset used in this study is available at: https://data.mendeley.com/datasets/nd64mvxtym/1 Code and data are available at GitHub: https://github.com/arfff527/or_llm_agent_OR_scheduling.

Code availability

Code and data are available at GitHub: https://github.com/arfff527/or_llm_agent_OR_scheduling.

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Chu Tang and Qu Wei contributed equally to this work and should be considered as co-first authors. Chu Tang: original draft, writing, validation, formal analysis, data curation. Qu Wei: original draft, writing, validation, formal analysis, data curation. Haitao Liu: writing, validation, formal analysis. Zekun Jiang: validation, formal analysis. Xiaobo Zhou: validation, formal analysis. Yu Cao: formal analysis, Data curation. Kang Li: writing—review & editing, supervision, conceptualization.

Competing interests

The authors declare no competing interests.

Additional information

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